MILITARY SPECIFICATION

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, CROSSLINKED, ELASTOMERIC POLYOLEFIN, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

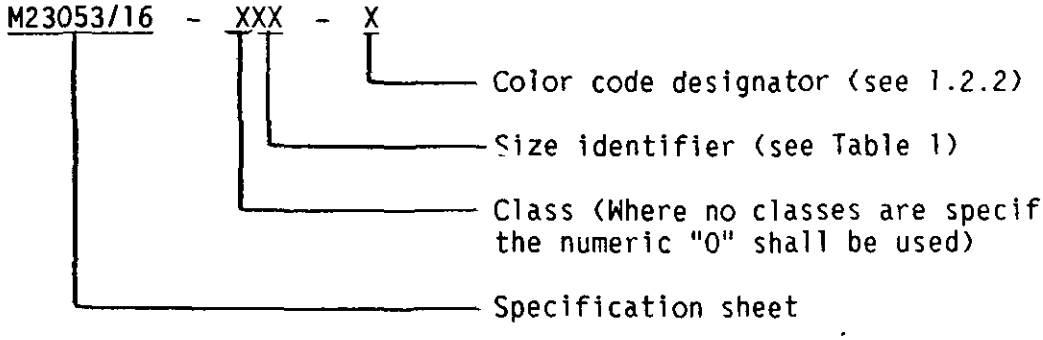
REQUIREMENTS:

Continuous Operating temperature range: -75°C (-103°F) to +120°C (+248°F).

Color: The sleeving shall be furnished in a black color that conforms to Class II of MIL-STD-104 (see 1.2.2 and 3.3.1.5).

Longitudinal change: ± 10 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



Example: Black, 1.000 inch (25.4 mm) as supplied ID sleeving shall be identified as M23053/16-005-0.

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

Military part number 2/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness, nominal
M23053/16-001-0	.250(6.35)	.125(3.18)	.035 (.889)
M23053/16-002-0	.375(9.53)	.187(4.75)	.040 (1.02)
M23053/16-003-0	.500(12.7)	.250(6.35)	.048 (1.23)
M23053/16-004-0	.750(19.1)	.375(9.53)	.057 (1.45)
M23053/16-005-0	1.000(25.4)	.500(12.7)	.070 (1.78)
M23053/16-006-0	1.500(38.1)	.750(19.1)	.095 (2.41)
M23053/16-007-0	2.000(50.8)	1.000(25.4)	.110 (2.79)
M23053/16-008-0	3.000(76.2)	1.500(38.1)	.125 (3.18)
M23053/16-009-0	4.000(101.6)	2.000(50.8)	.140 (3.56)

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ The color code identified is the standard acquisition color.

Unrestricted shrinkage procedures: Test method 4.6.5; 175° ± 2°C (347° ± 4°F) for 10 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Low temperature flexibility	No cracks	4.6.7.1 -65° ± 2°C (-85° ± 4°F)
Heat shock	No cracks, flowing or dripping	4.6.8 200° ± 2°C (392° ± 4°F)
Restricted shrinkage	No cracks	4.6.6.1.1 150° ± 2°C (302° ± 4°F)
Voltage withstand	Pass	4.6.6.2

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	1,700(11.7)	4.6.14 ASTM D 412
Ultimate elongation, percent, minimum	250	4.6.14 ASTM D 412
Tensile stress at 100 percent elongation, psi (MPa), maximum	1,500(10.4)	4.6.13.2 ASTM D 412
Dielectric strength, volts/mil (Kv/mm), minimum	300 (11.9)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm, minimum	1×10^{10}	4.6.2 ASTM D 876
Specific gravity, maximum	1.5	4.6.2 ASTM D 792
Corrosion	No corrosion or pitting	4.6.10.1 121° ± 2°C (250° ± 4°F); 4.6.10.2 135° ± 2°C (275° ± 4°F); for 16 hours
Water absorption, percent, maximum	2.0	4.6.2 ASTM D 570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 150° ± 2°C (302° ± 4°F), 168 hours
Tensile strength, psi (MPa), minimum	1,500(10.3)	
Ultimate elongation, percent, minimum	200	

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Fluid resistance, properties after:		4.6.11 2/
Tensile strength, psi (MPa), minimum	1,500(10.4)	
Ultimate elongation, percent, minimum	200	
Dielectric strength, volts/mil (Kv/mm), minimum	200(7.9)	
Flammability	Self-extinguishing within one minute and no more than 25 percent of indicator flag burned or charred	4.6.15.1
Fungus resistance	Rating less than 1	4.6.2 ASTM G 21

1/ Unless otherwise specified, requirements, test procedures and conditions are for all classes.

2/ In addition to the fluids specified in 4.6.11, the sleeving shall also be immersed in diesel fuel conforming to MIL-F-46162.

Shelf life conditions: Supplier shall certify to storage at 45°C (113°F) for 2 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable flexible elastomeric sleeving is intended for use on heavy duty cables or harness systems, such as ground support which see high levels of physical abuse and exposure to fuels and oils, coupled with high and low temperature extremes.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5970-0577-15)

Review activities:
Army - MI, MU
Navy - EC
Air Force - 80
DLA - GS

User activities:
Army - ME
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-23053 /16A	INSULATION SLEEVING, ELECTRICAL / HEAT SHRINKABLE GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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